

# The Hyperon Nucleon Interaction in a Low Energy Effective Field Theory

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**In collaboration with**

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**Domenico Logoteta** *(INFN)*

**Laura Elisa Marcucci** *(University of Pisa, INFN)*



**26th European Conference on Few  
Body Problems in Physics,  
Prague**

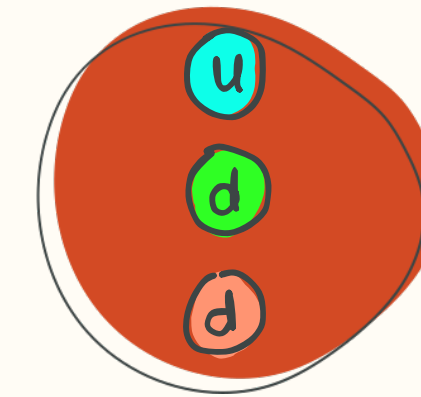


# Aim of the project

Develop a potential model for the YN interaction

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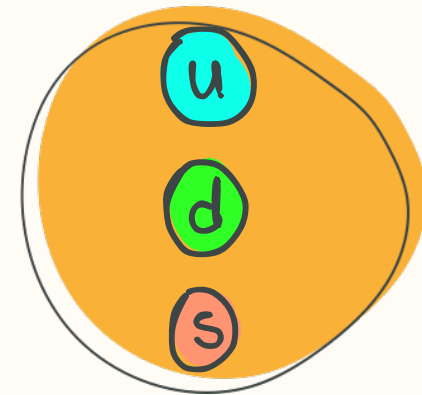
Develop a potential model for the  $YN$  interaction



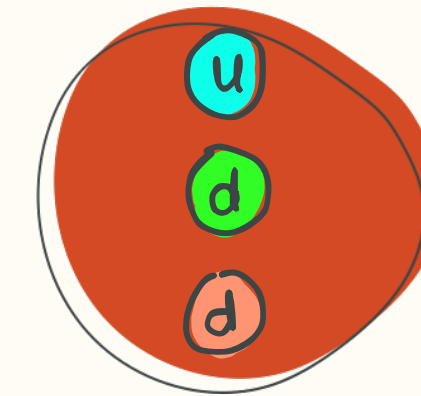
*Nucleon (p)*

# Aim of the project

Develop a potential model for the  $YN$  interaction



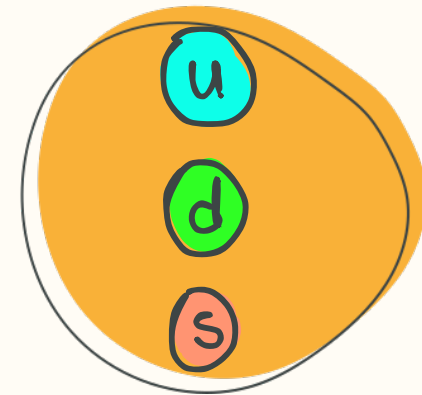
*Hyperon ( $\Lambda$ )*



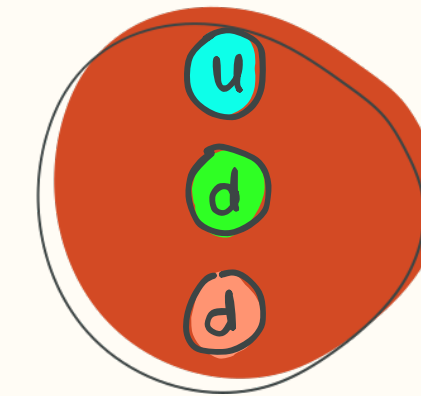
*Nucleon ( $p$ )*

# Aim of the project

Develop a potential model for the  $\bar{Y}N$  interaction



*Hyperon ( $\Lambda$ )*



*Nucleon (p)*

- *Baryons*
- *At least one strange quark*
- *Fermions*
- *Lifetimes  $\sim 10^{-10} s$*

# Introduction and Motivations

Develop a potential model for the YN interaction

Local model

Contact EFT approach

# Introduction and Motivations

## Develop a potential model for the YN interaction

### Local model



Depending only on relative distance between the particles



### Contact EFT approach

# Introduction and Motivations

## Develop a potential model for the YN interaction

### Local model



Depending only on relative distance between the particles



### Contact EFT approach



Includes constants that are fixed through a fitting procedure

$$V = \sum_i \underbrace{C_i}_{\text{FIT}} O_i$$

# Introduction and Motivations

Develop a potential model for the YN interaction

Local model

Contact EFT approach



# Introduction and Motivations

Develop a potential model for the YN interaction

Local model

Contact EFT approach

→ *Simple model* ←



*Easy to handle numerically  
(Also suitable for computationally  
expensive tasks)*

# Introduction and Motivations

Few Body

Applications

Many Body

# Introduction and Motivations

Few Body

*Hypernuclei and correlation function studies*

Applications

Many Body



# Introduction and Motivations

## Few Body

*Hypernuclei and correlation function studies*

## Applications

## Many Body

*Neutron Stars equation of state and hyperon puzzle*



# Introduction and Motivations

## Few Body

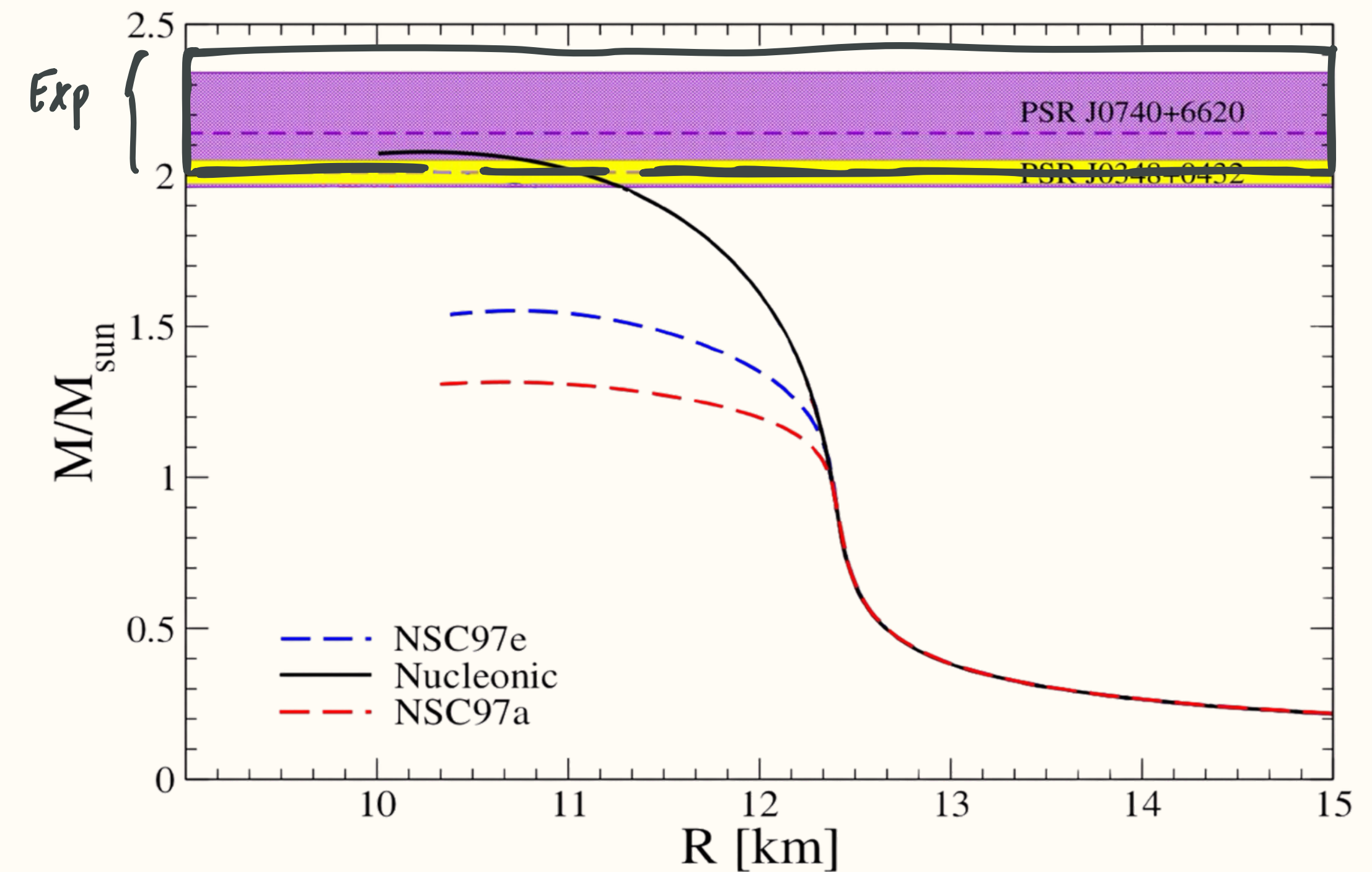
*Hypernuclei and correlation function studies*



## Applications

## Many Body

*Neutron Stars equation of state and hyperon puzzle*



Adapted from D. Logoteta, I. Vidaña & I. Bombaci (2019), [arXiv:1906.11722](https://arxiv.org/abs/1906.11722)

# Introduction and Motivations

## Few Body

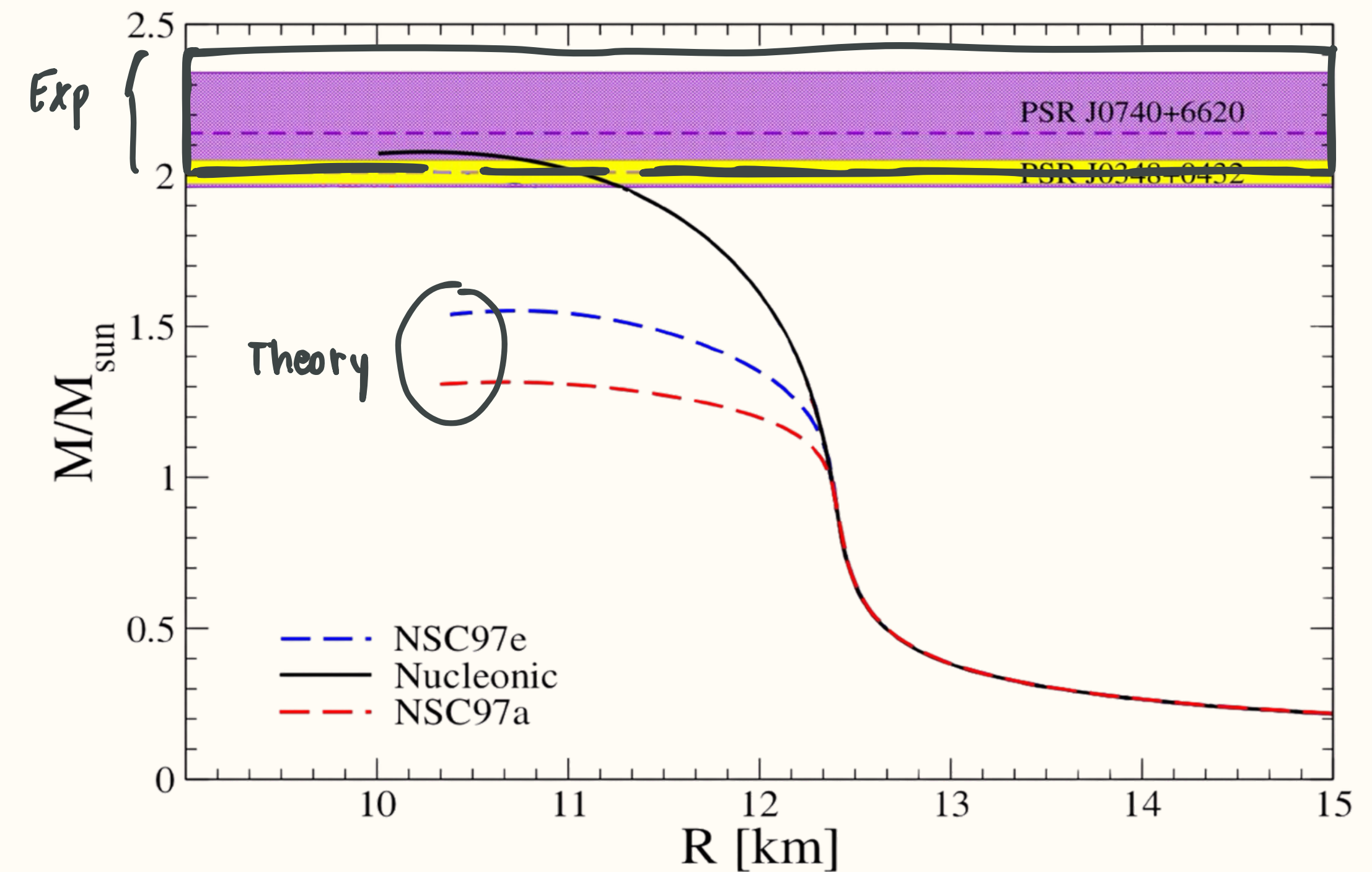
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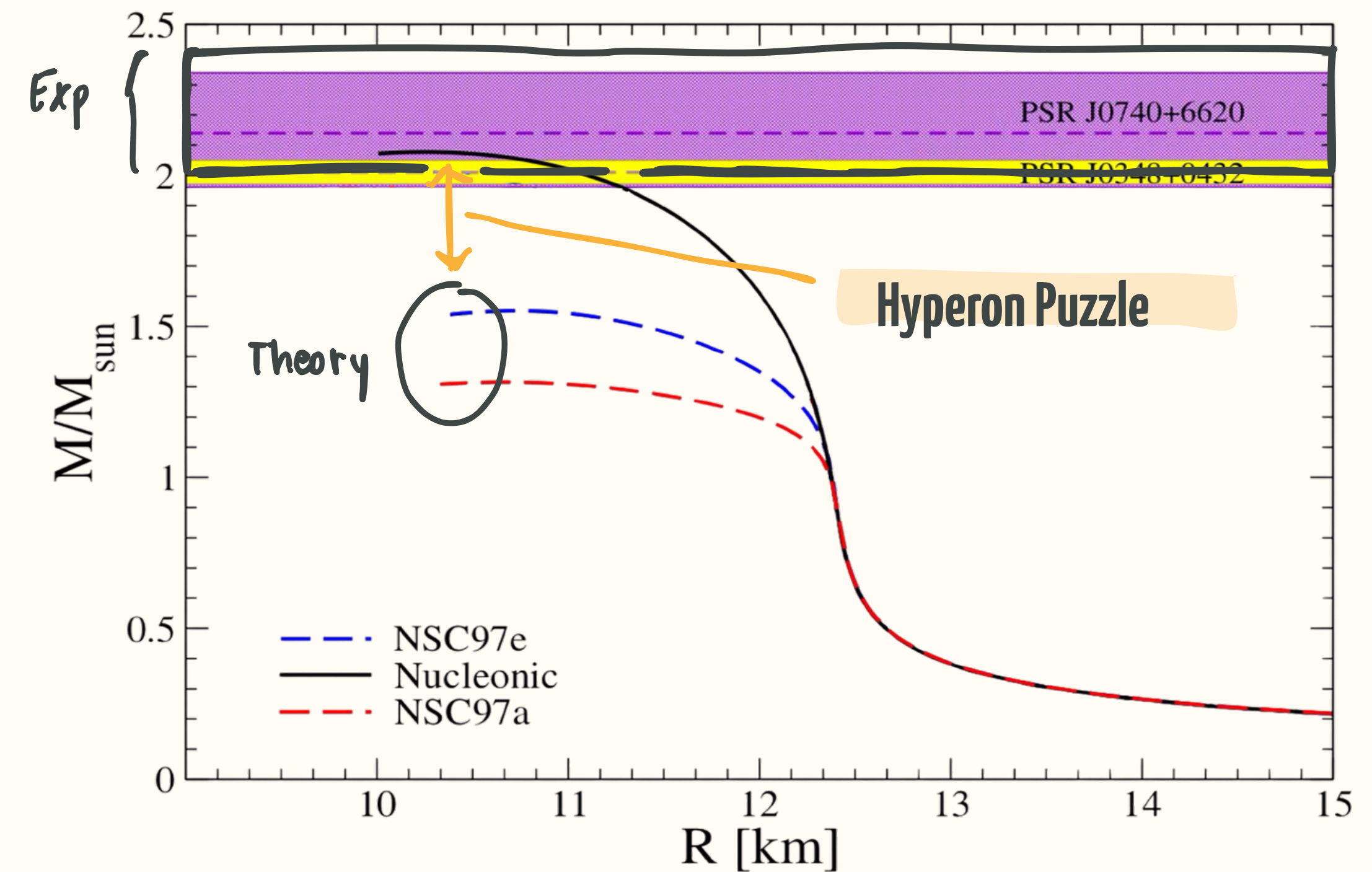
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## Many Body

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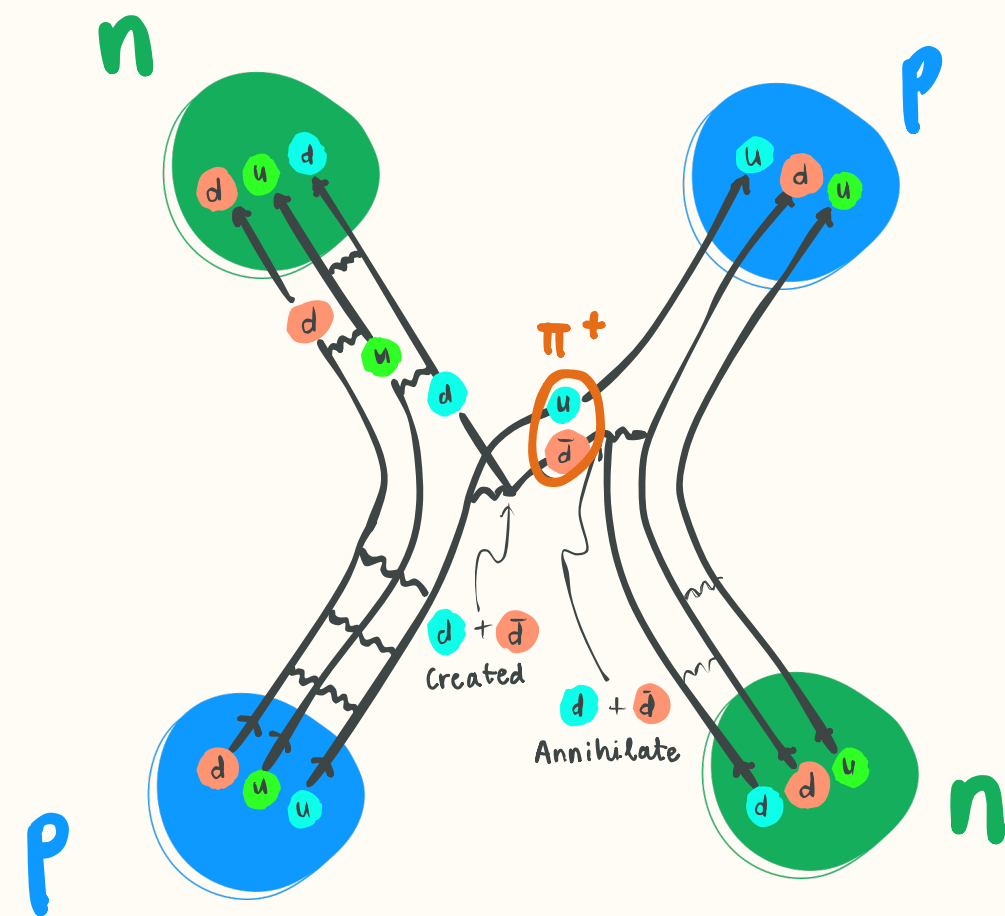
# Contact EFT in a nutshell

QCD

– dof: quarks, gluons

Energy

$\sim 1 \text{ GeV}$

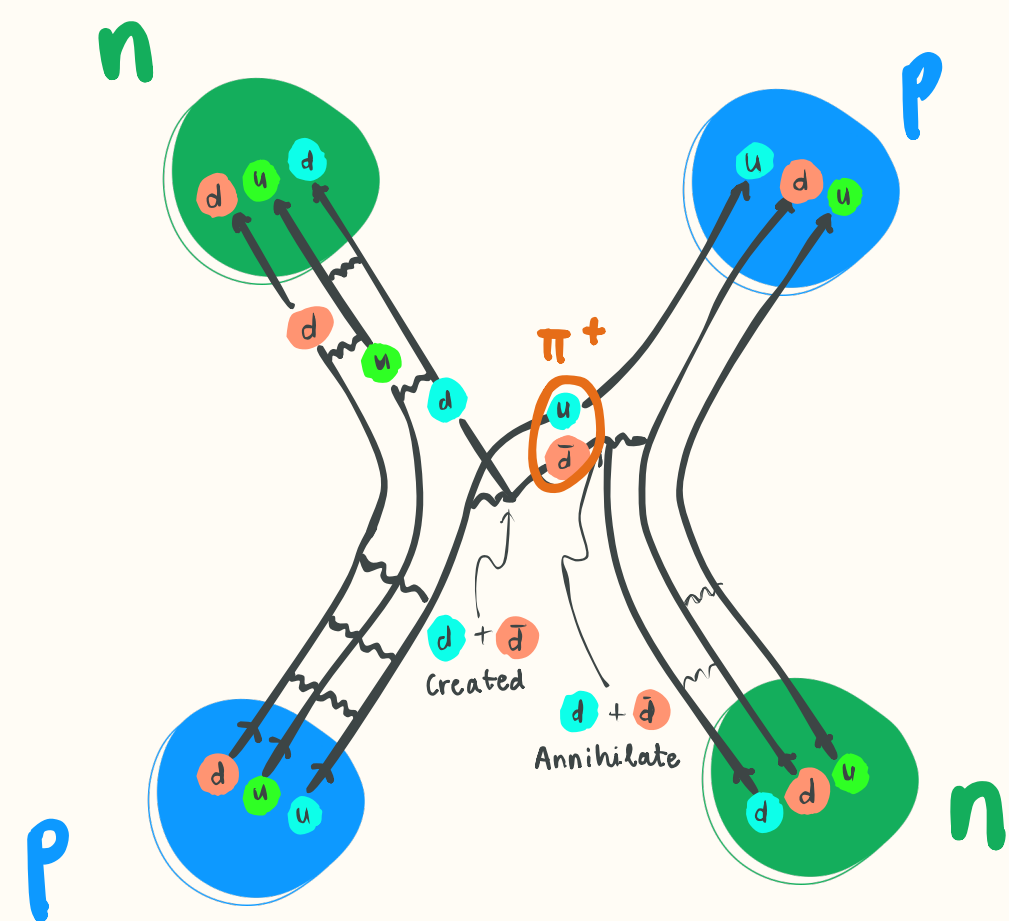


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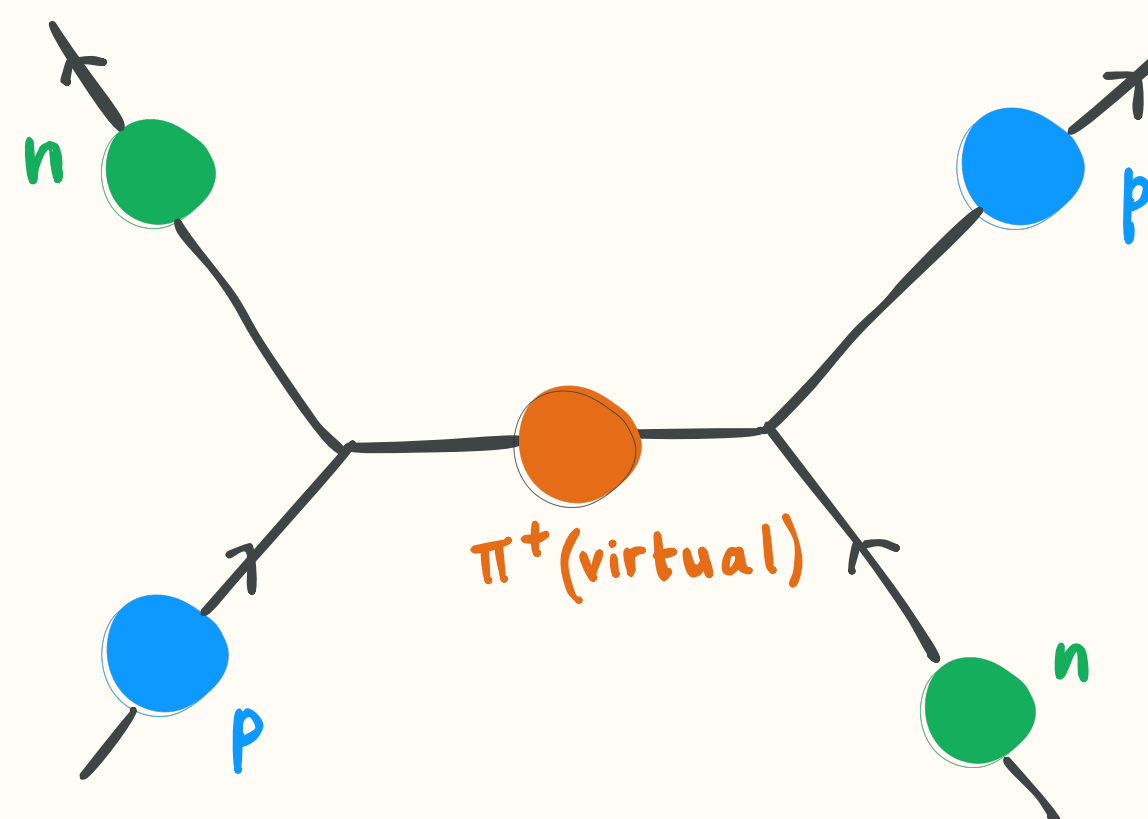
~ 1 GeV



$\chi$ EFT

– dof: nucleons, pions

~ 100 MeV

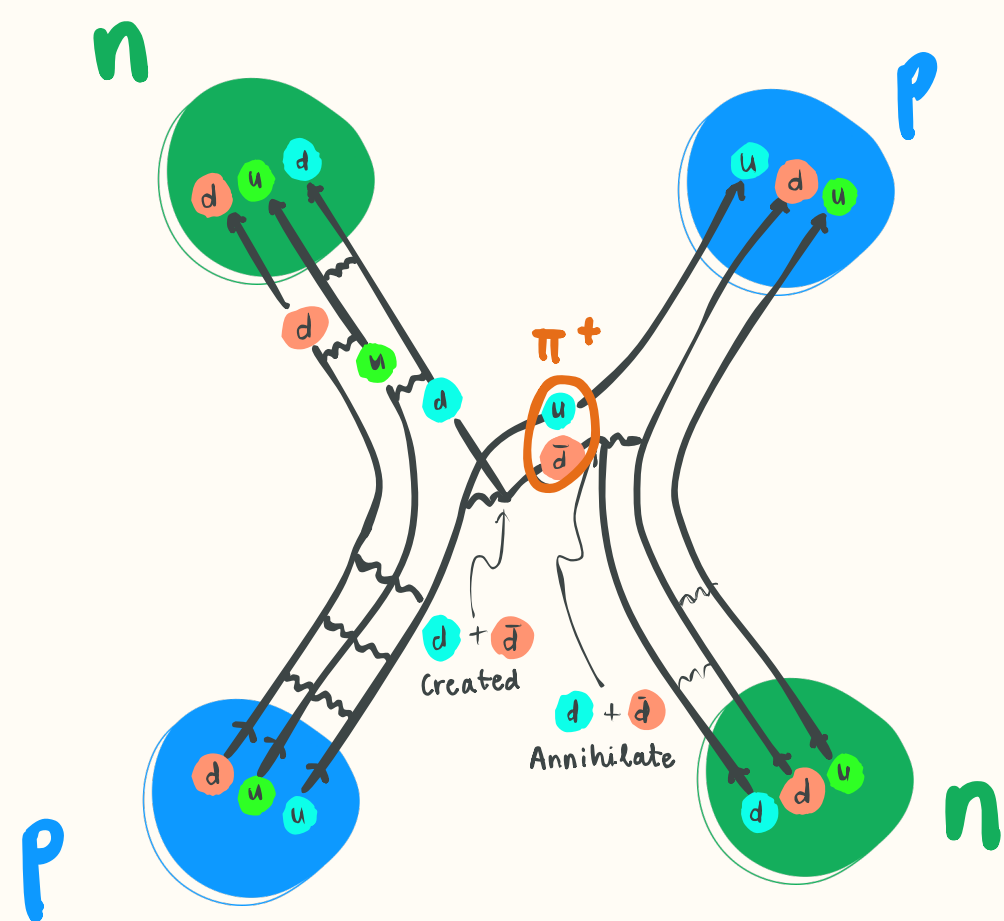


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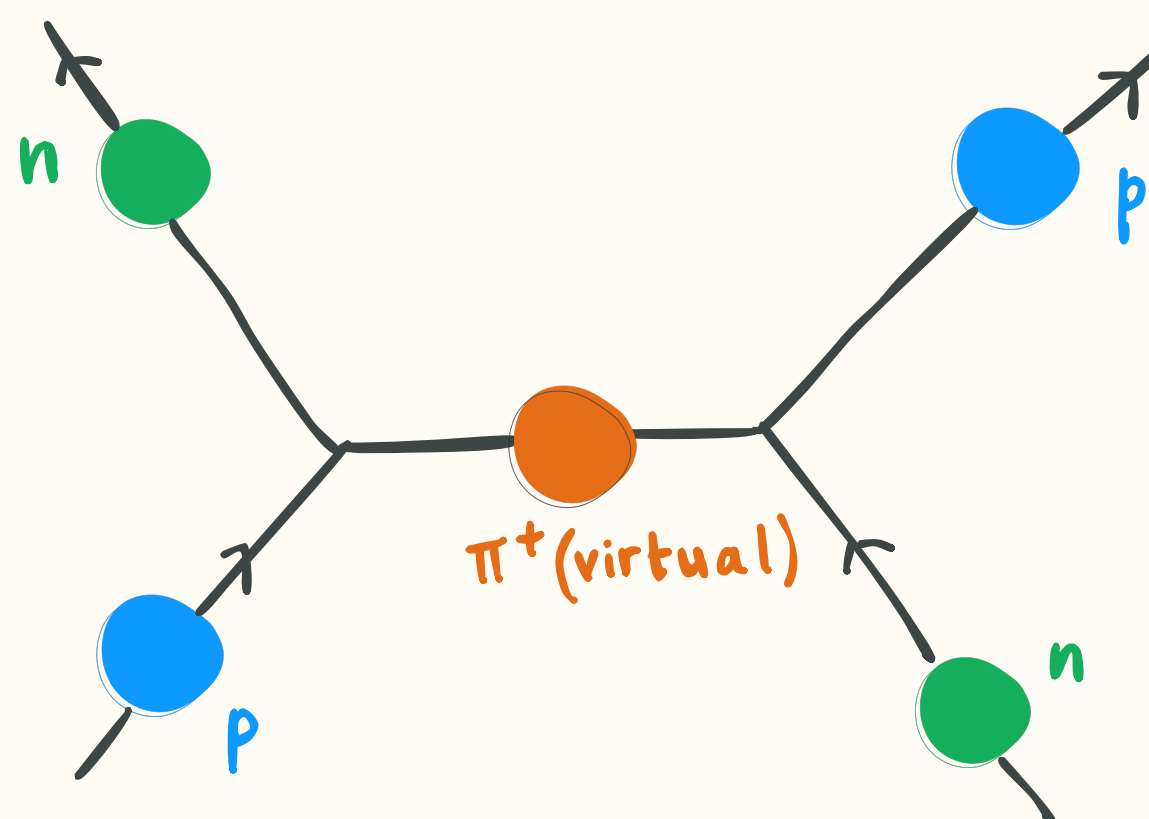
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$\chi$ EFT

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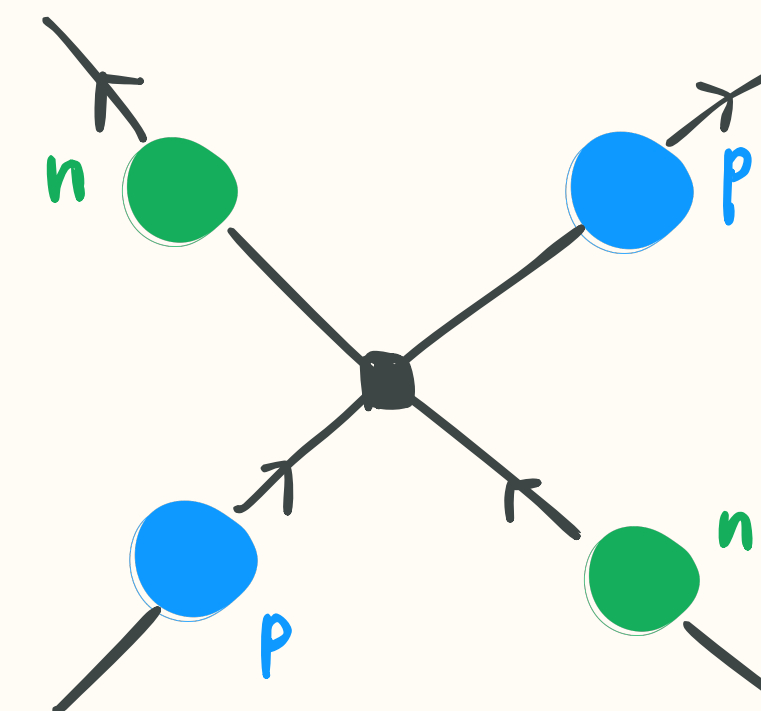
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~~$\pi$~~ EFT

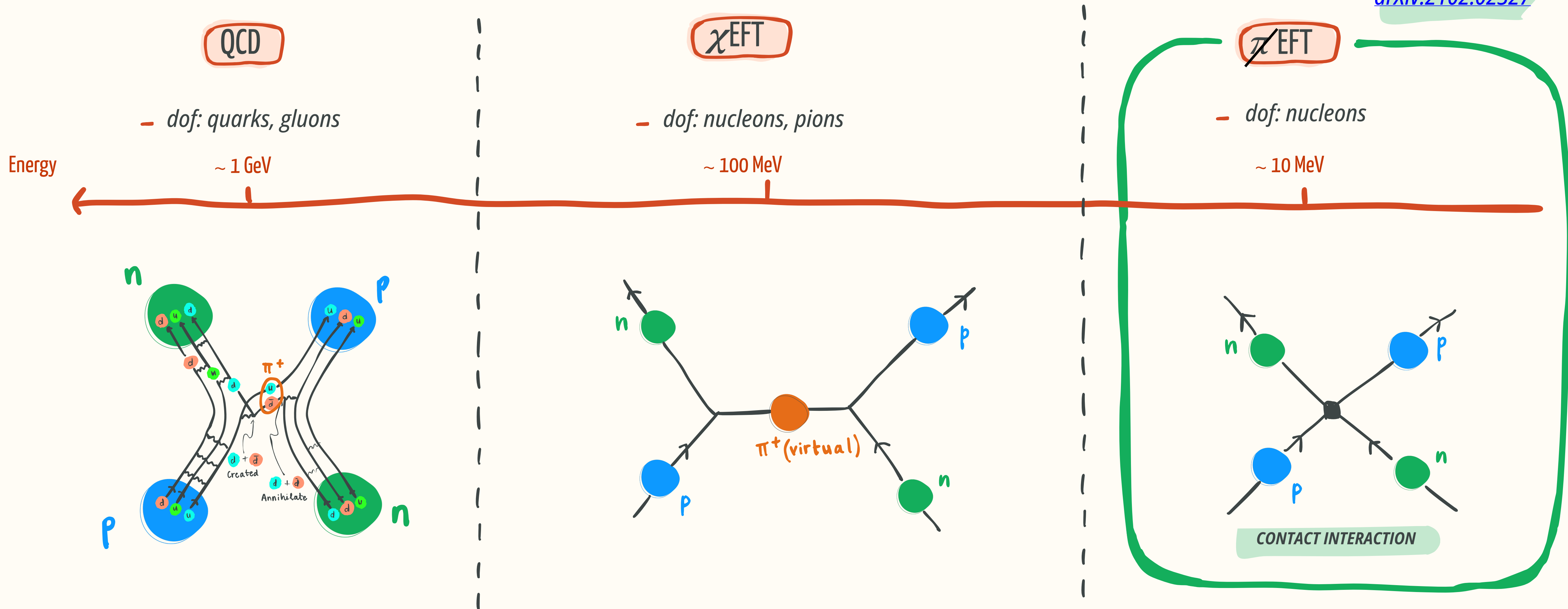
– dof: nucleons

~ 10 MeV



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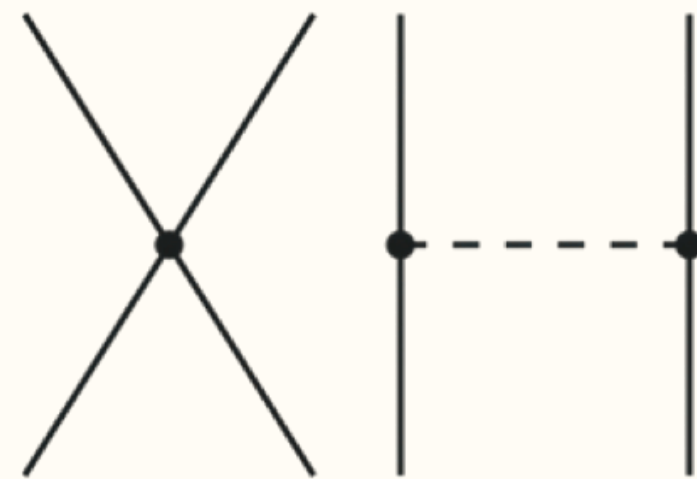
Schiavilla et al. (2021)  
[arXiv:2102.02327](https://arxiv.org/abs/2102.02327)



# Contact pionless approach

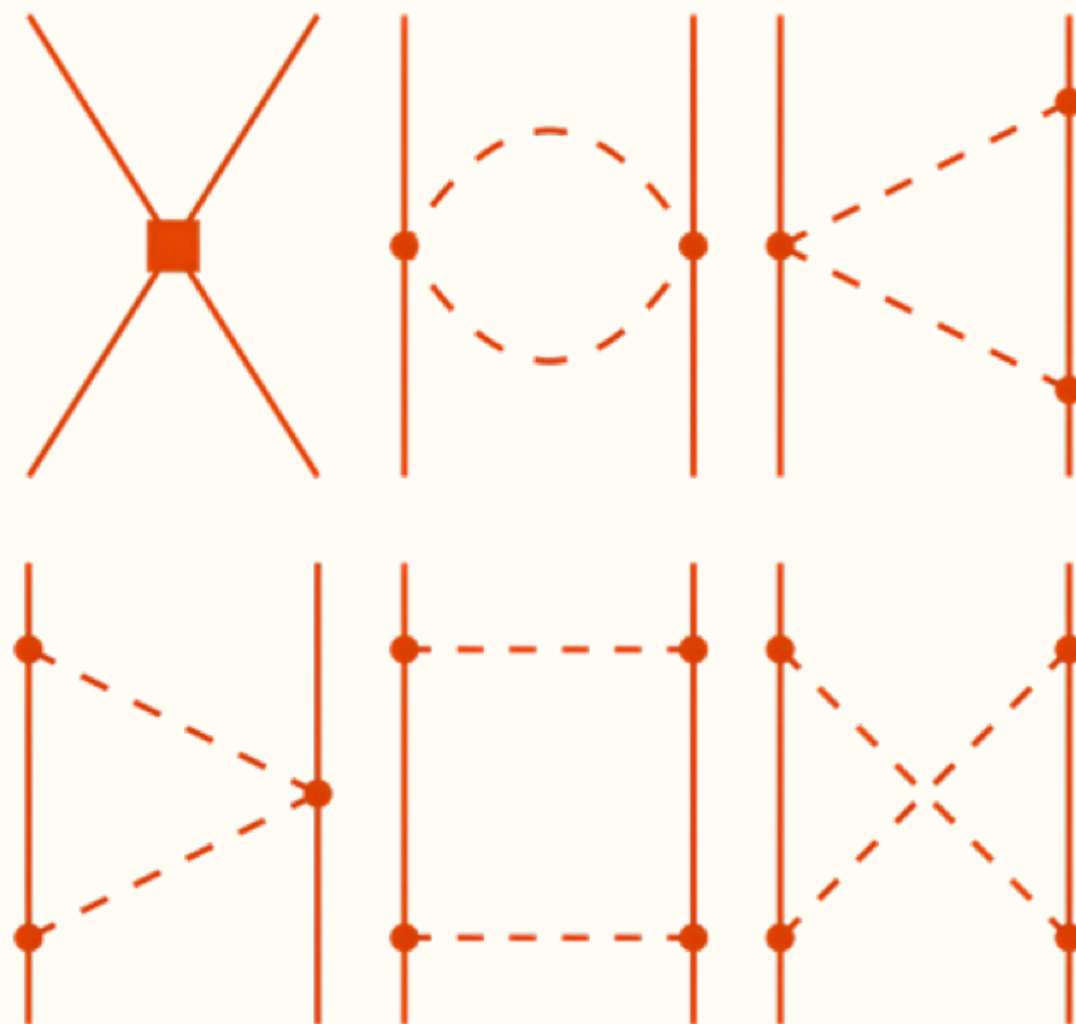
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**LO**  
 $(Q/\Lambda_\chi)^0$



- *Keep same power counting as  $\chi$ EFT*

**NLO**  
 $(Q/\Lambda_\chi)^2$

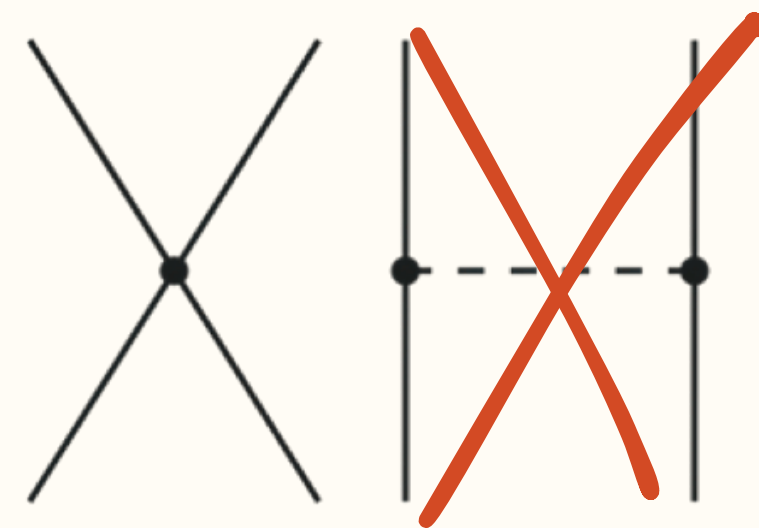


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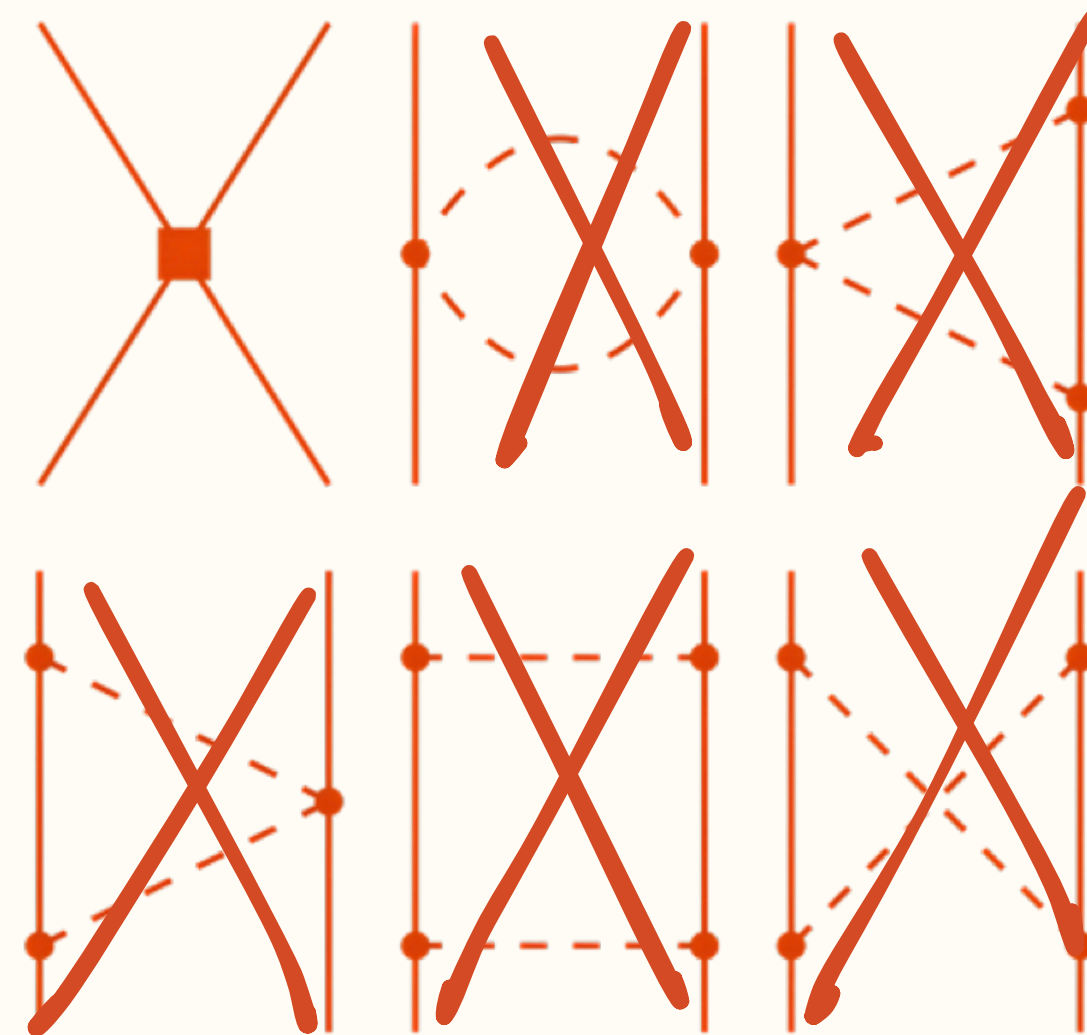
## 2N Force

LO  
 $(Q/\Lambda_\chi)^0$



- *Keep same power counting as  $\chi$ EFT*

NLO  
 $(Q/\Lambda_\chi)^2$



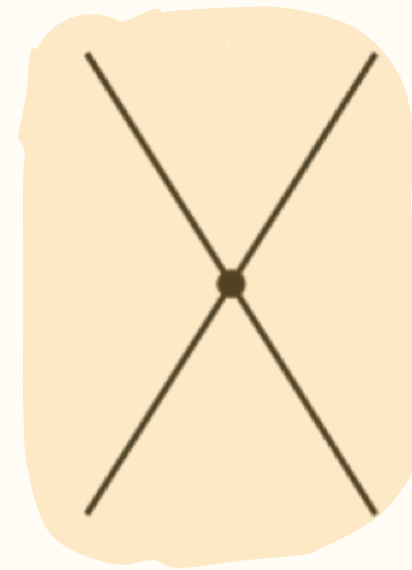
- X** *Remove diagrams that involve pion exchanges*

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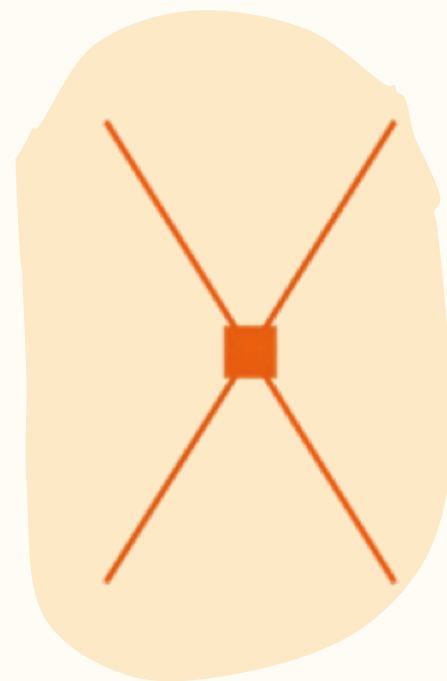
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## 2N Force

LO  
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NLO  
 $(Q/\Lambda_\chi)^2$



Only contact terms

- *Keep same power counting as  $\chi$ EFT*

- ✗ *Remove diagrams that involve pion exchanges*

# $\Lambda N$ interaction in coordinate space

- *Schiavilla et al. (2021), [arXiv:2102.02327](#) : contact pionless approach (for NN interaction), coordinate space*
- *Haidenbauer et al. (2023), [arXiv:2301.00722](#):  $YN$  interaction with  $\chi$ EFT approach, momentum space*

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$$\begin{aligned}
 V_{\Lambda N}^{LO} &= [C_S + C_T(\sigma_\Lambda \cdot \sigma_N)]F(r) \\
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 &\quad - C_3 \frac{F^{(1)}(r)}{r} \mathbf{L} \cdot \mathbf{S} - C_4 \left( F^{(2)}(r) - \frac{1}{r}F^{(1)}(r) \right) S_{\Lambda N}(\hat{\mathbf{r}}) - C_5 \frac{F^{(1)}(r)}{r} \mathbf{L} \cdot \mathbf{D}
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 \end{aligned}$$

$F(r) = \frac{1}{\pi^{3/2} R_0^3} \exp\left(-\frac{r^2}{R_0^2}\right)$   
 $R_0 \in [1, 2.5] \text{ fm}$   
 Cutoff parameter

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$F(r)$  first derivative

$F(r)$  second derivative

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 \end{aligned}$$

$\mathbf{D} = \frac{(\sigma_\Lambda - \sigma_N)}{2}$

$\mathbf{S} = \frac{(\sigma_\Lambda + \sigma_N)}{2}$

$S_{\Lambda N}(\hat{\mathbf{r}}) = 3\sigma_\Lambda \cdot \hat{\mathbf{r}}\sigma_N \cdot \hat{\mathbf{r}} - \sigma_\Lambda \cdot \sigma_N$

*orbital angular momentum*

# $\Lambda N$ interaction in coordinate space

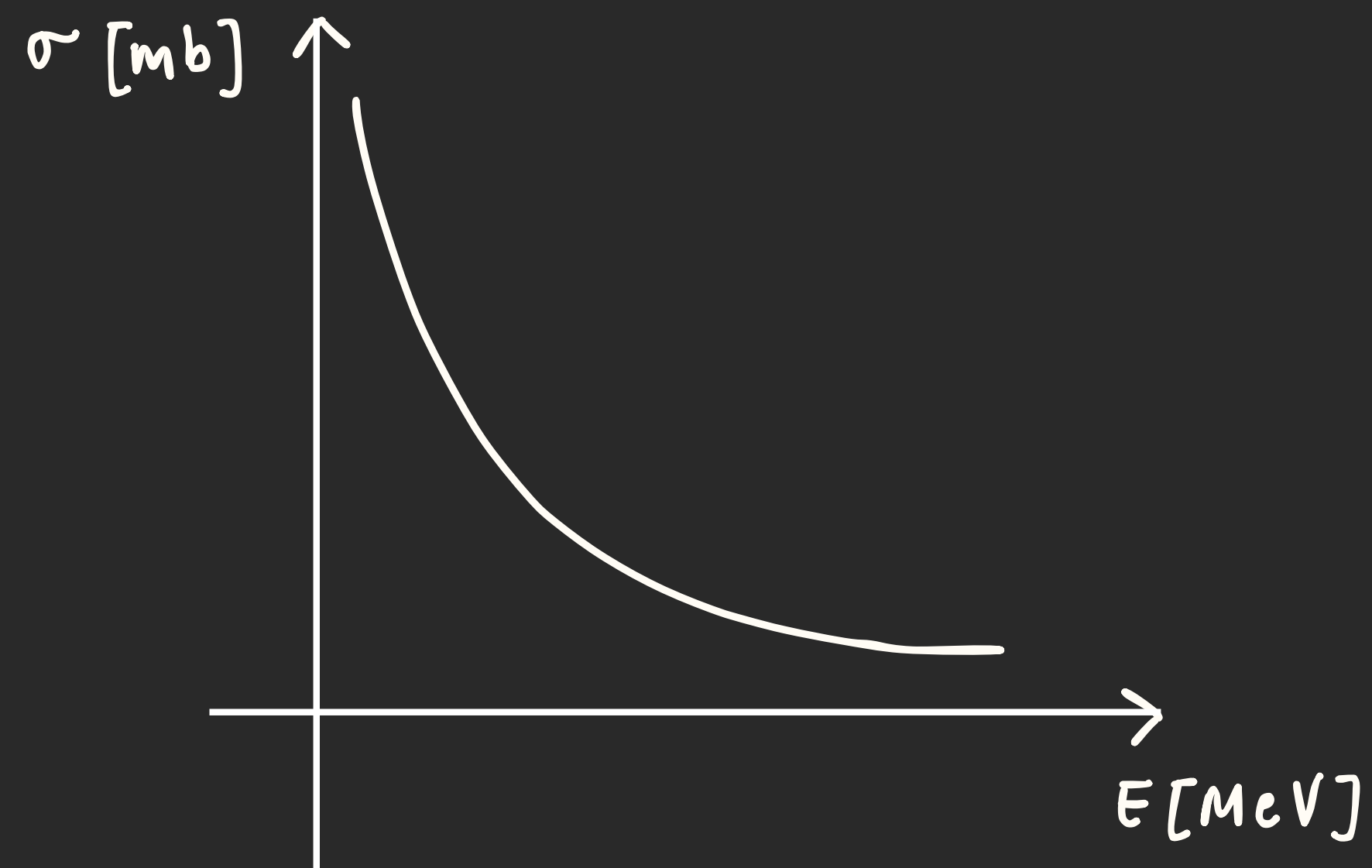
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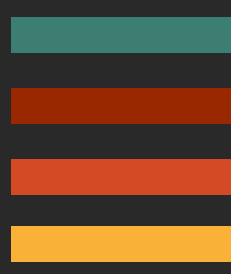
Low Energy Constants  
(LECs)



*fixed through fitting to experimental data  
( $\Lambda p$  elastic scattering cross section)*



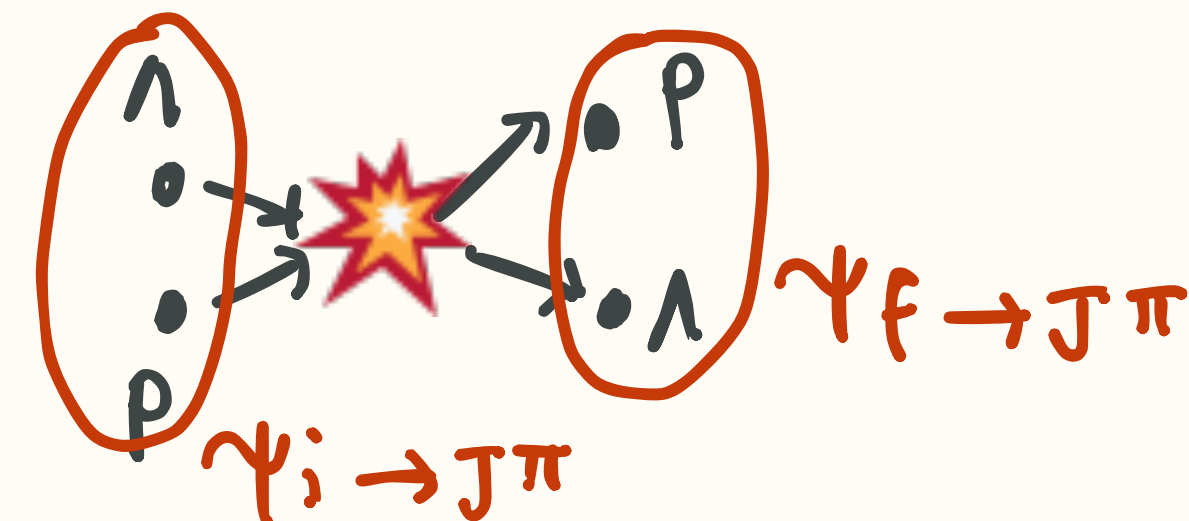
 Cross Section

Calculation 



# Cross section calculation

Available experimental data:  
 $\Delta p$  elastic scattering cross section



Initial and final scattering states  
 expanded in partial waves

| $J\pi$ | $\alpha$ | $L$ | $S$ | $^{2S+1}L_J$ |
|--------|----------|-----|-----|--------------|
| $0^+$  | 1        | 0   | 0   | $^1S_0$      |
| $0^-$  | 1        | 1   | 1   | $^3P_0$      |
| $1^+$  | 1        | 0   | 1   | $^3S_1$      |
|        | 2        | 2   | 1   | $^3D_1$      |
| $1^-$  | 1        | 1   | 0   | $^1P_1$      |
|        | 2        | 1   | 1   | $^3P_1$      |

# Cross section calculation

Available experimental data:

$\Delta p$  elastic scattering cross section

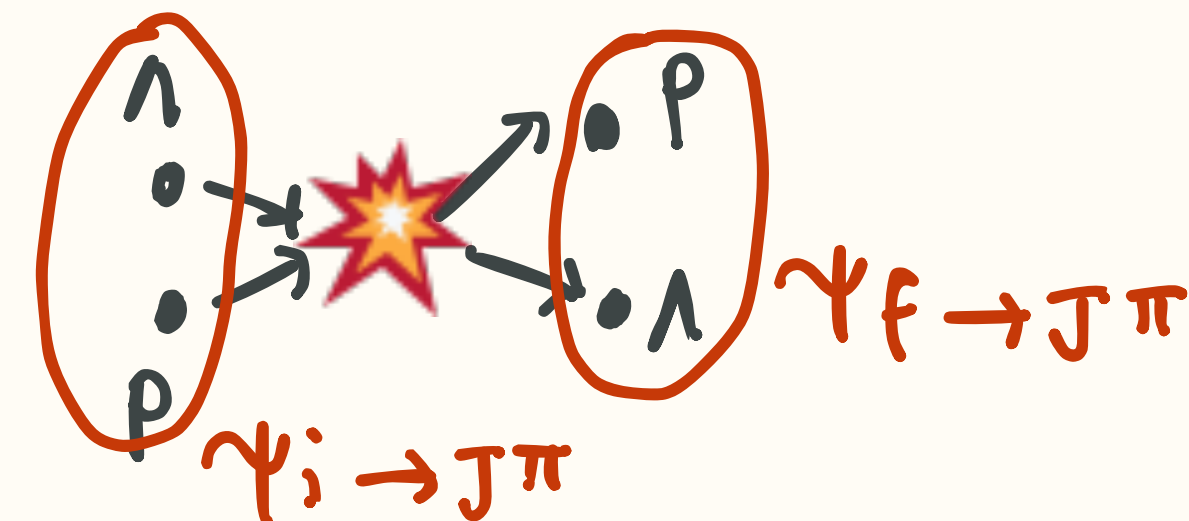


**Total unpolarized cross section**

$$\sigma = \frac{4\pi}{q^2} \sum_{J, \alpha, \beta} |T_{\alpha\beta}|^2 \frac{(2J+1)}{(2s_{\Lambda}+1)(2s_N+1)}$$

**T-Matrix**

(depends on LECs, computed with **Kohn variational principle**)



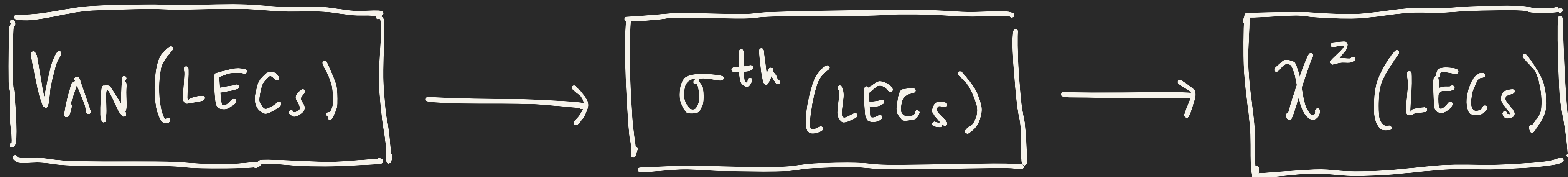
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|           | 2        | 2   | 1   | $^3D_1$    |
| $1^-$     | 1        | 1   | 0   | $^1P_1$    |
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# Fitting

# procedure



# Fitting procedure

$\chi^2$  function with a constraint on scattering length  $\longrightarrow a_j = \pm \lim_{q \rightarrow 0} \frac{\sin \delta_j}{q}$

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$$\chi^2 = \sum_i \frac{[\sigma_i^{th}(C_S, C_T, C_1, \dots, C_5) - \sigma_i^{exp}]^2}{err(\sigma_i^{exp})^2} + \sum_{j=s,t} \frac{[a_j^{th}(C_S, C_T, C_1, \dots, C_5) - a_j^{exp}]^2}{err(a_j^{exp})^2}$$

Cross section experimental data from  
Radboud University, NN online archive

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$^1S_0$        $^3S_1$

Cross section experimental data from  
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"Experimental" data for scattering length  
from Mihaylov et al. (2024), [arXiv:2312.16970](https://arxiv.org/abs/2312.16970)

$$a_s^{exp}, a_t^{exp} = (2.1, 1.56) \text{ to } (3.34, 1.18) \text{ fm}$$

# Fitting procedure

- TAOPOUNDERS, from *PETSc* + *MPI* to parallelize
- Fit performed at LO and NLO separately for different cutoff values  $R_0$  !

$$V_{\Lambda N}^{LO} = [C_S + C_T(\sigma_\Lambda \cdot \sigma_N)]F(r)$$

2 free parameters

$$E_{CM} \leq 15 \text{ MeV}$$

$$J^\pi = 0^+, 1^+$$

mainly S-wave

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$$E_{CM} \leq 80 \text{ MeV}$$

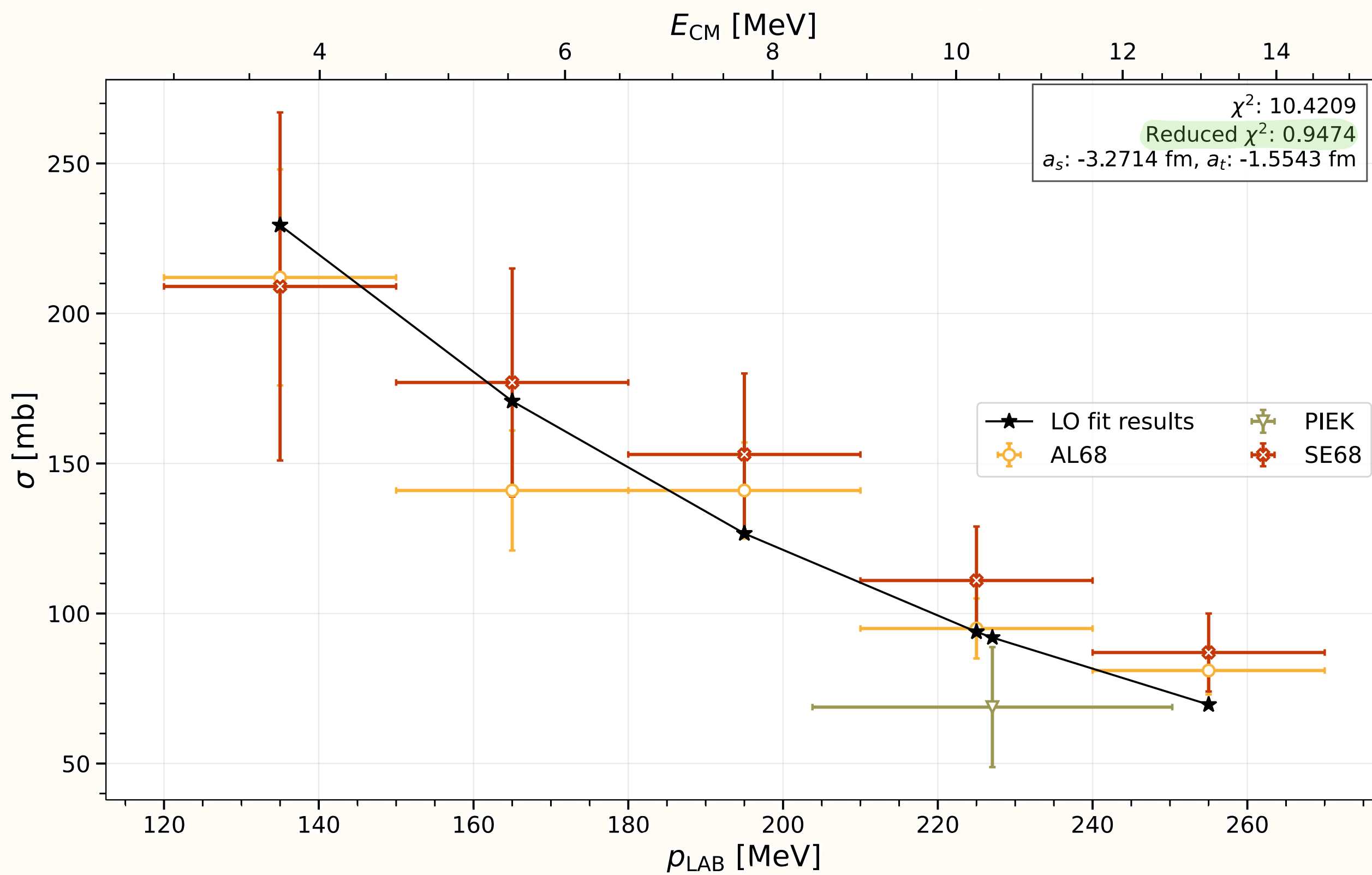
$$J^\pi = 0^\pm, 1^\pm$$

S-wave and P-wave

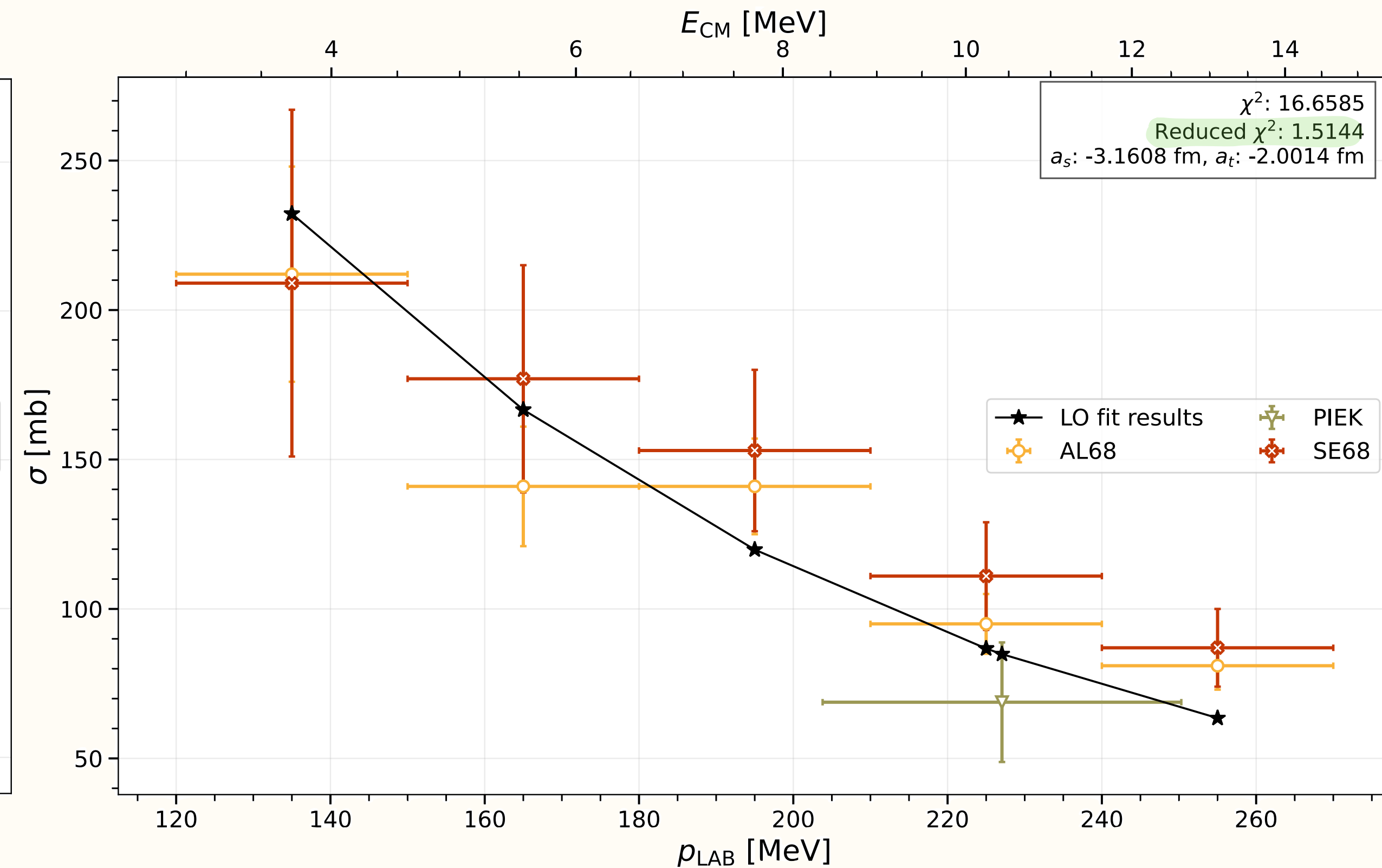
7 free parameters

# Results LO

$R_0 = 1 \text{ fm}$

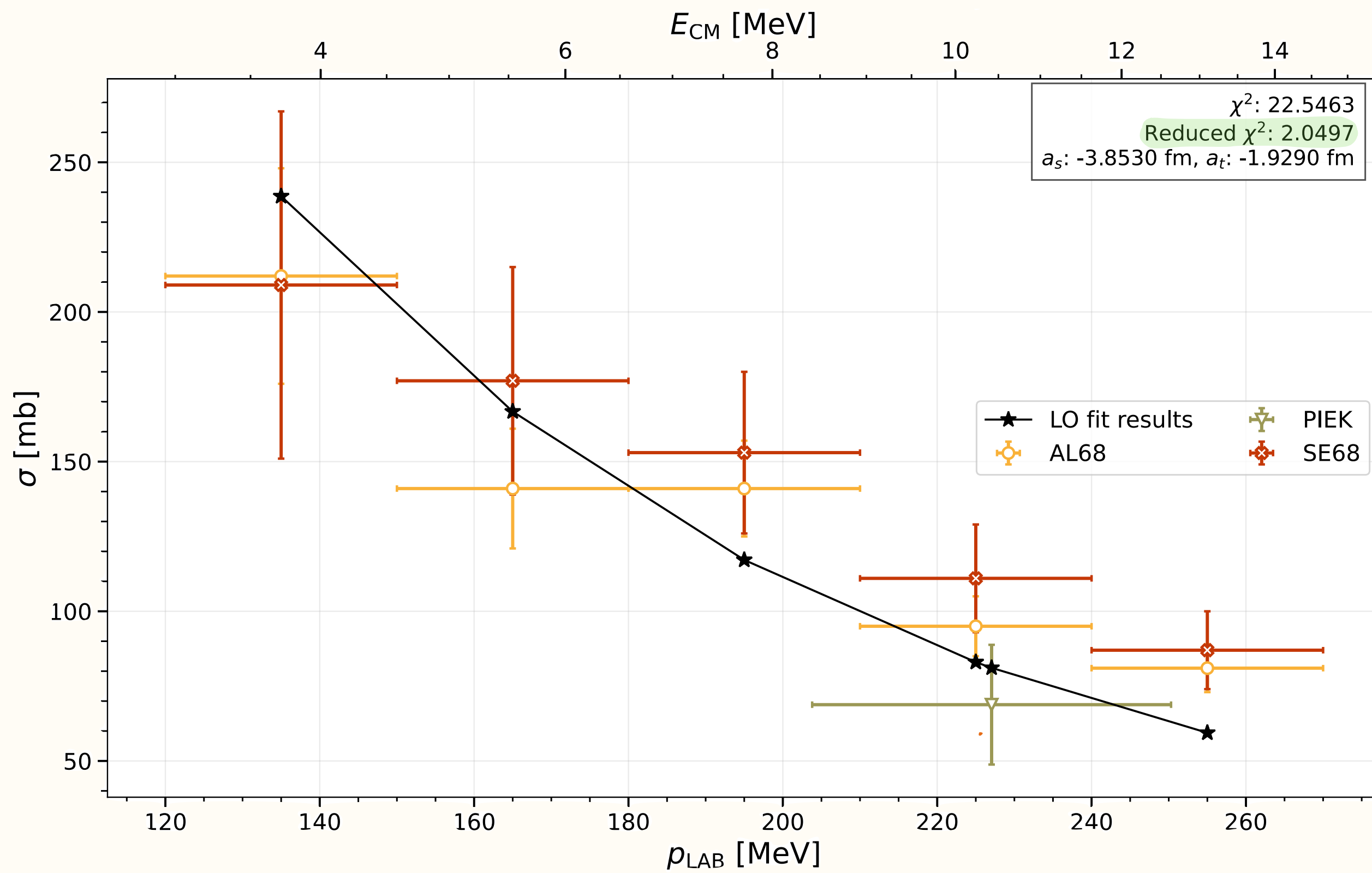


$R_0 = 1.5 \text{ fm}$

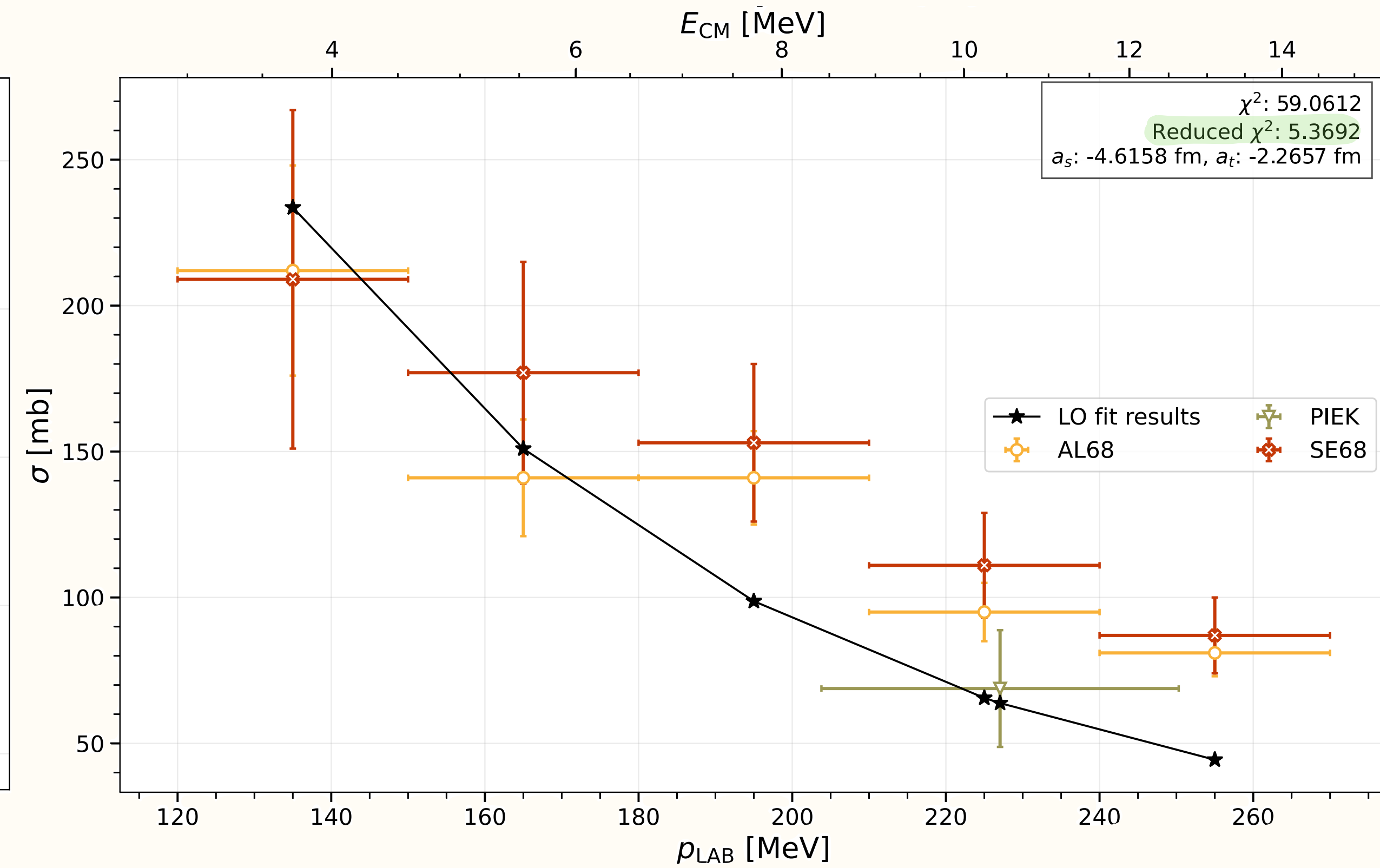


# Results LO

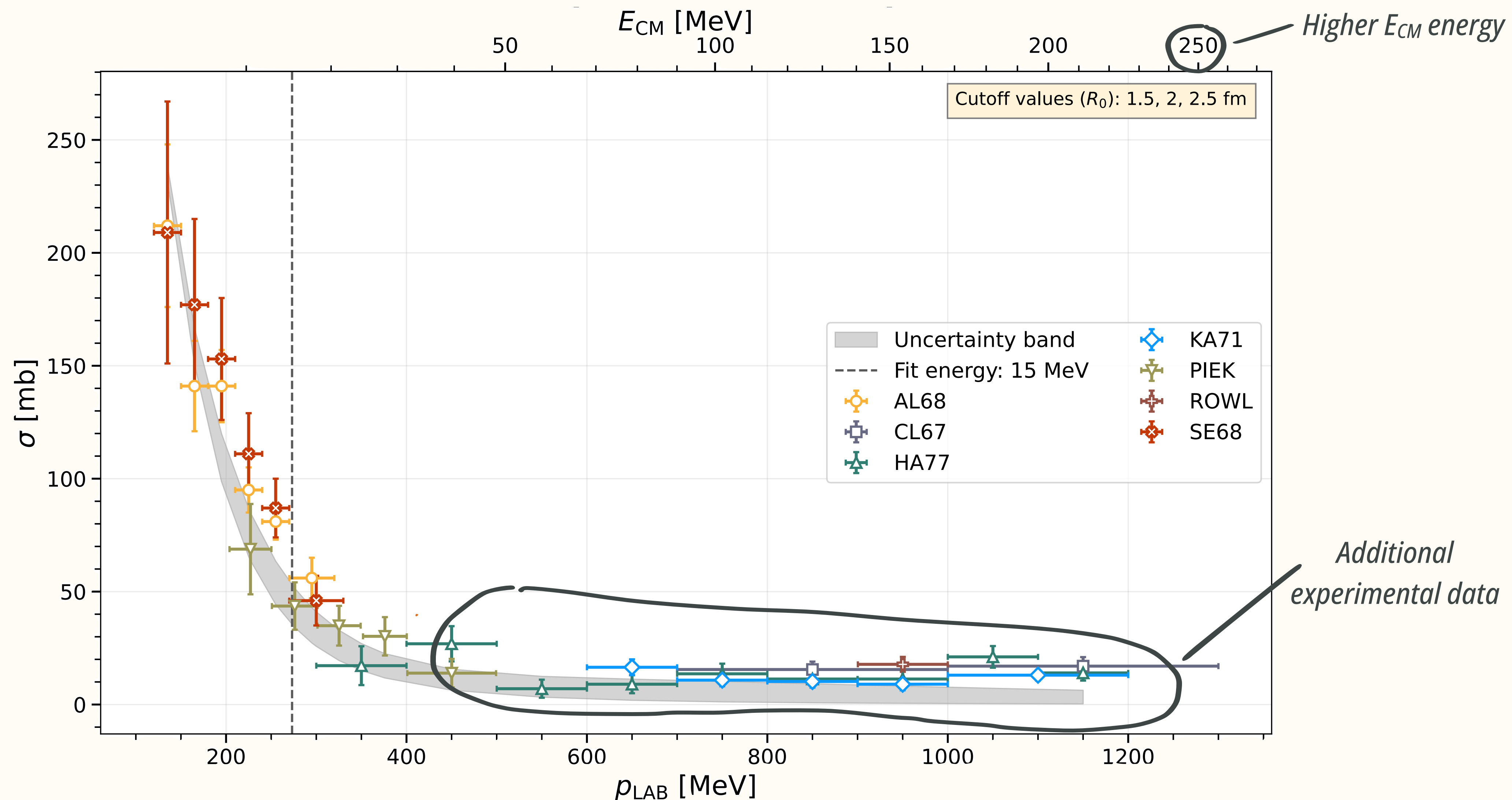
$R_0 = 2 \text{ fm}$



$R_0 = 2.5 \text{ fm}$

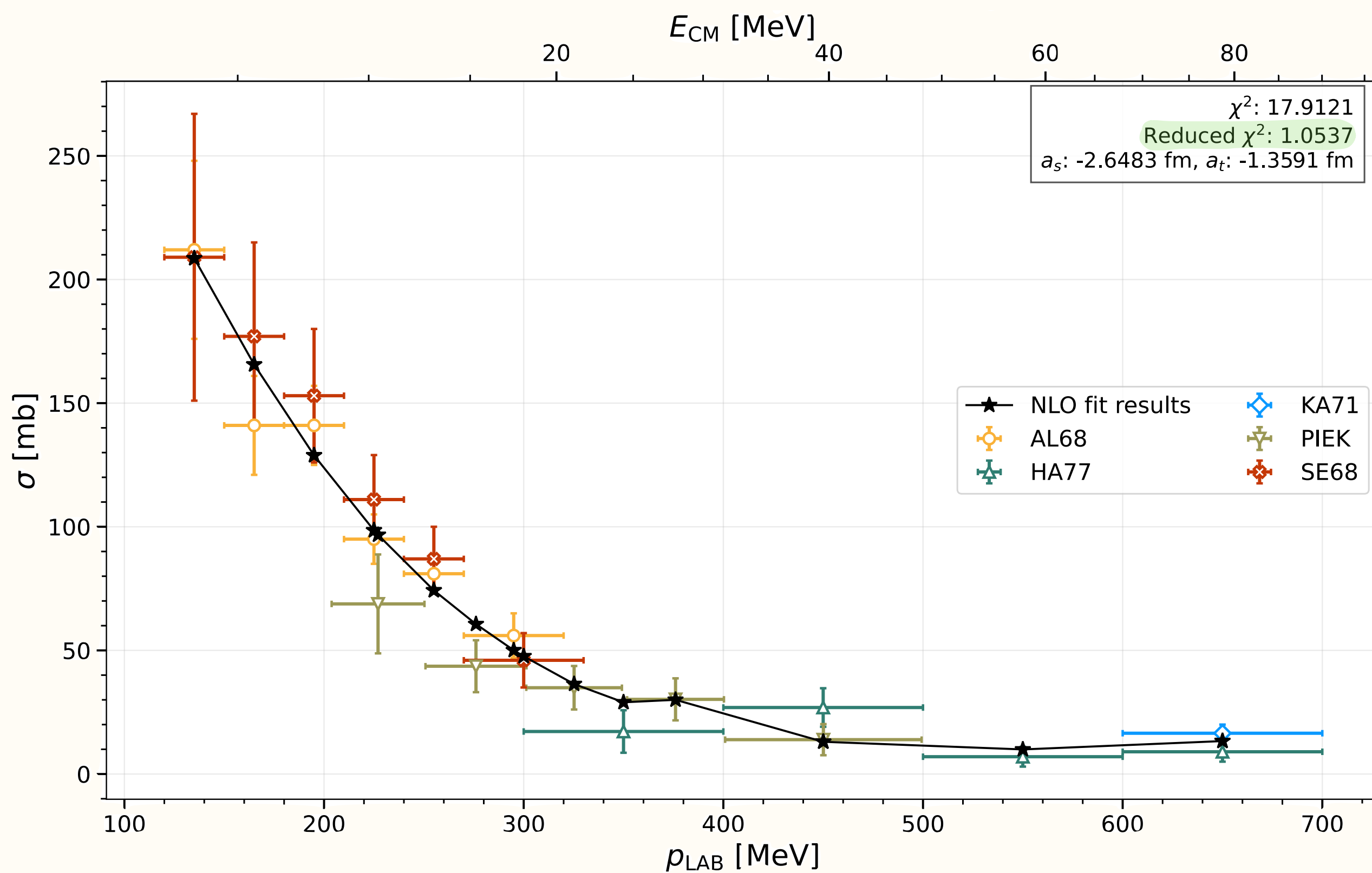


# Results L0 - high energy predictions

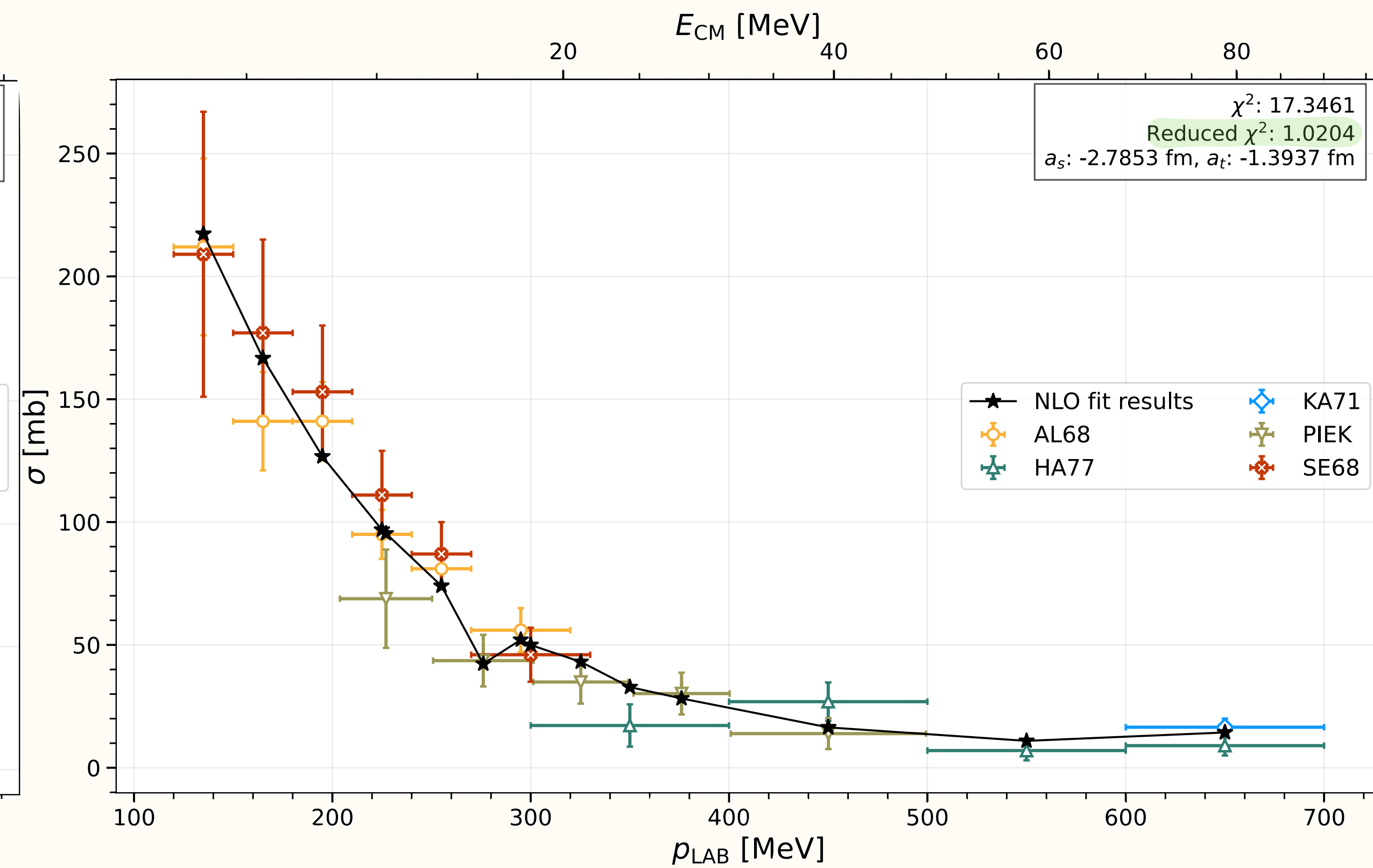


# Results NLO

$R_0 = 1 \text{ fm}$

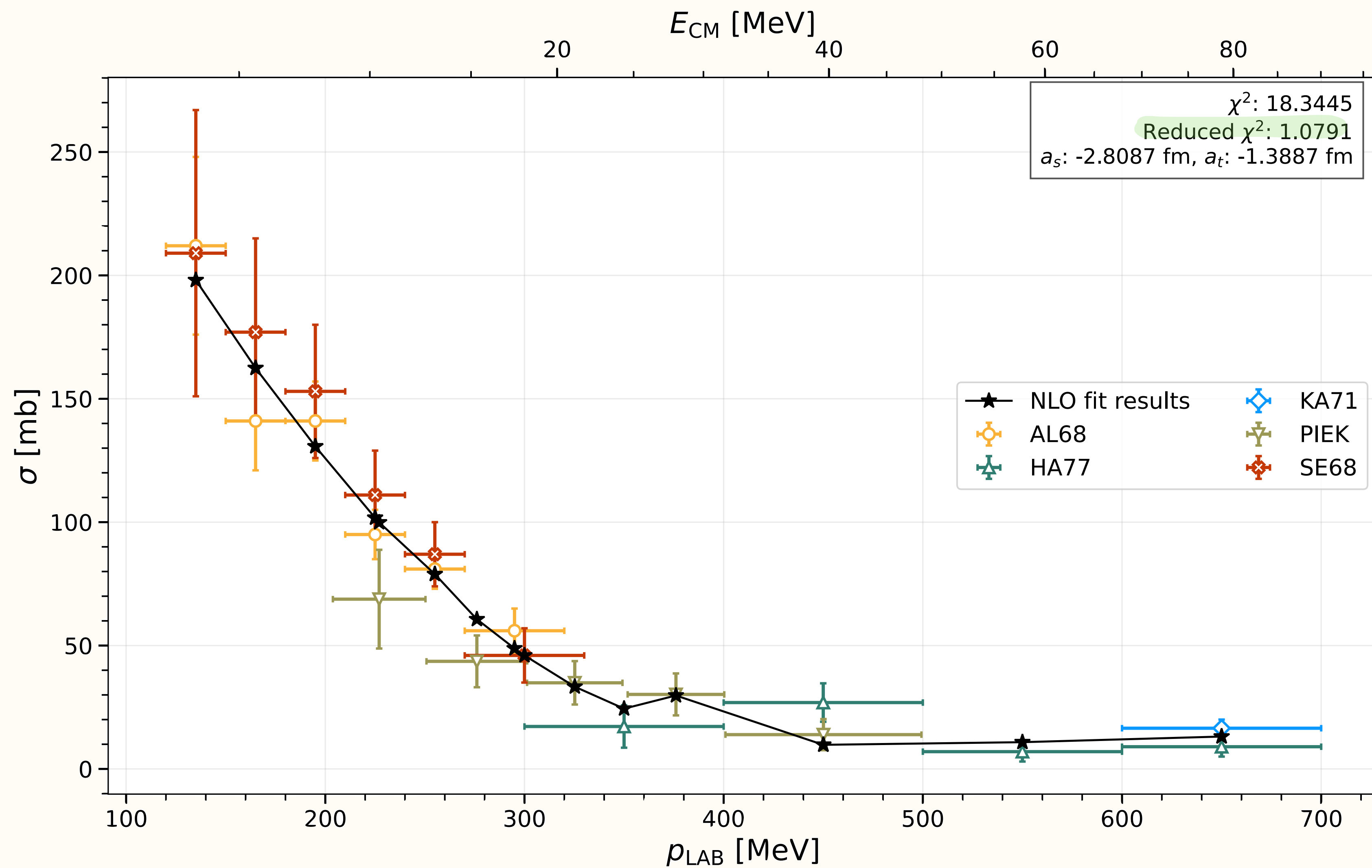


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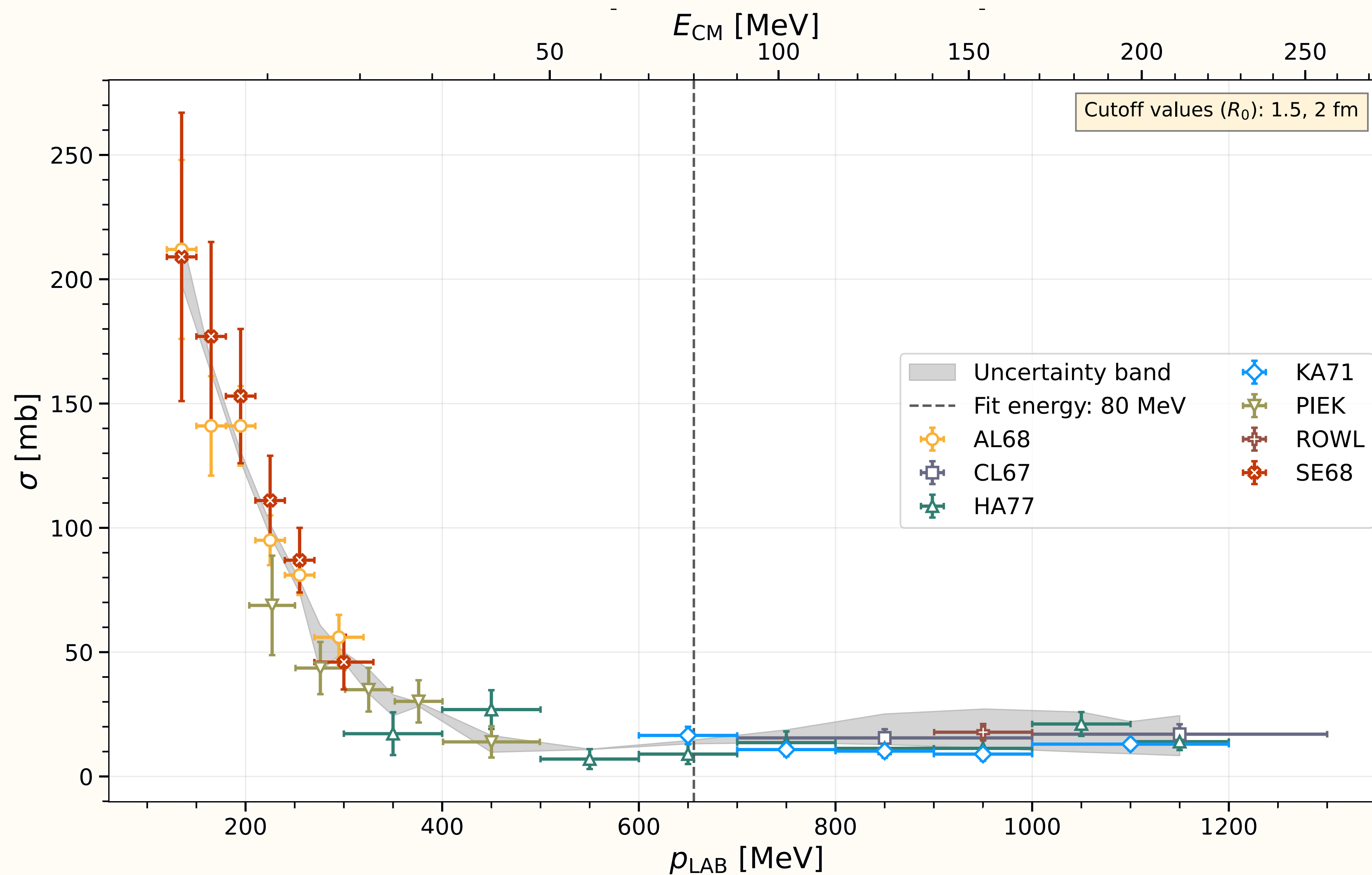


# Results NLO

$R_0 = 2 \text{ fm}$



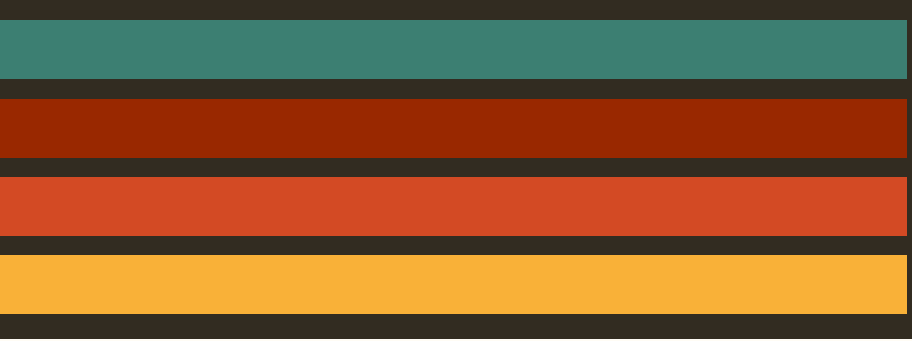
# Results NLO - high energy predictions





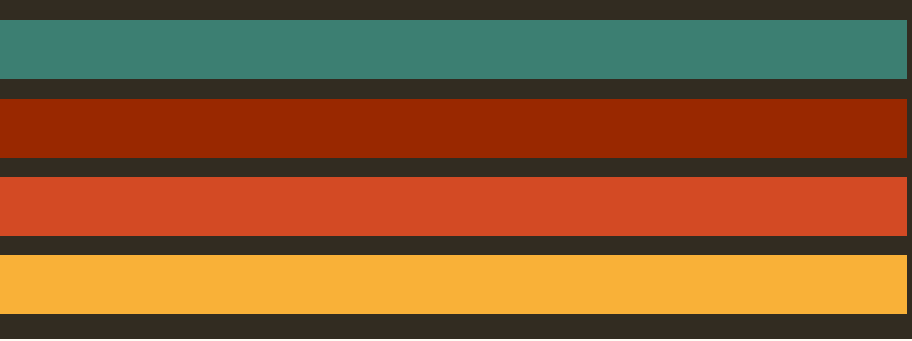
# Applications to light hypernuclei





# ${}^3_{\Lambda}\text{H}$ binding energy

- *Schiavilla et al. (2021), [arXiv:2102.02327](https://arxiv.org/abs/2102.02327) : contact pionless approach (for NN interaction), up to NLO*
- *Current work (for  $\Lambda N$  interaction), up to NLO*



# ${}^3_{\Lambda}\text{H}$ binding energy

- *Schiavilla et al. (2021), [arXiv:2102.02327](#) : contact pionless approach (for NN interaction), up to NLO*
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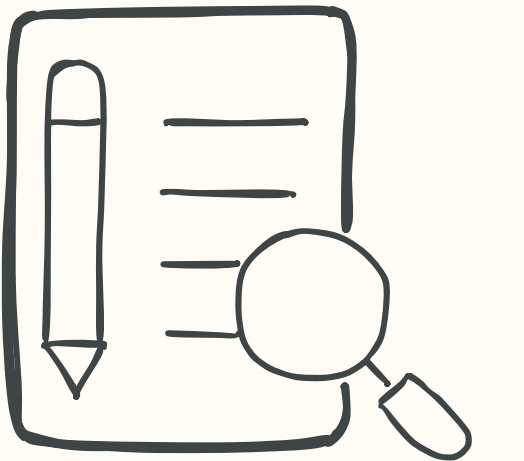
${}^3\text{H}$  binding energy  $\pi\text{EFT}$  LO results,  
NLO is work in progress

| Model | NSHH    | HH      |
|-------|---------|---------|
| $a$   | −10.706 | −10.705 |
| $b$   | −8.463  | −8.463  |
| $c$   | −7.066  | −7.066  |
| $d$   | −6.137  | −6.136  |
| $o$   | −9.696  | −9.696  |

# Current work & future developments

*Developed a local contact potential model for the  $\Lambda p$  interaction up to NLO*

- *Sophisticated fitting procedure*
- *Compatibility with scattering data and scattering lengths*



SUMMARY

# Current work & future developments

*Developed a local contact potential model for the  $\Lambda p$  interaction up to NLO*

- *Sophisticated fitting procedure*
- *Compatibility with scattering data and scattering lengths*



*Applications to light hypernuclei*

*Develop a local  $\chi$ EFT potential model*

*In collaboration with  
**Maria Piarulli** (Washington University in St Louis)*

*Studies on Neutron Stars' Equation of State*

**>>**  
**Future**

**Margherita Sagina**  
*(University of Pisa, INFN)*

**Thank you**   
 **for your attention!**

# $\chi^2$ function

- *First attempt:*

$$\chi^2 = \sum_i \frac{[\sigma_i^{th}(C_S, C_T, C_1, \dots, C_5) - \sigma_i^{exp}]^2}{err(\sigma_i^{exp})^2}$$



Gives  $V_{\Lambda N} = 0$  at LO  
when total spin  $S=1$

- *Second attempt: constraint on scattering length*

$$\chi^2 = \sum_i \frac{[\sigma_i^{th}(C_S, C_T, C_1, \dots, C_5) - \sigma_i^{exp}]^2}{err(\sigma_i^{exp})^2} + \sum_{j=s,t} \frac{[a_j^{th}(C_S, C_T, C_1, \dots, C_5) - a_j^{exp}]^2}{err(a_j^{exp})^2}$$

✓ Solves  $S=1$   
issue

# $\Lambda N$ interaction - partial waves projection

- *Partial waves projection of the potential, in momentum space*

$$V(^1S_0) = 4\pi(C_S - 3C_T) + \pi(4C_1 + C_2 - 12C_3 - 3C_4 - 4C_6 - C_7)(p^2 + p'^2)$$

$$V(^3S_1) = 4\pi(C_S + C_T) + \pi\frac{3}{2}(12C_1 + 3C_2 + 12C_3 + 3C_4 + 4C_6 + C_7)(p^2 + p'^2)$$

LO LECs

!

*We can't separate singlet and triplet contribution*



*need for scattering lengths*

# Fitting procedure

*Semi classical approach to compute  $p_{LAB}$  threshold for  $L0$*

$$\begin{array}{lcl} \ell = 1 & \nearrow & \ell \hbar = p_{LAB} b \\ \text{to exclude P-waves and higher-} & & \downarrow \\ \text{order partial waves} & & b \sim 1 \text{ fm} \\ & & \hbar \cdot c = 197.33 \text{ MeV} \cdot \text{fm} \end{array} \quad \left. \vphantom{\begin{array}{l} \ell \hbar = p_{LAB} b \\ b \sim 1 \text{ fm} \\ \hbar \cdot c = 197.33 \text{ MeV} \cdot \text{fm} \end{array}} \right\} \begin{array}{l} p_{LAB} \sim 200 \text{ MeV} \\ \Rightarrow E_{CM} \sim 15 \text{ MeV} \end{array}$$